

KELLOG, Russian Federation

WMO: 237740

Lat: **62.4892N**Lon: **86.2814E**Elev: **60**StdP: **100.60**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.5	-43.1	-49.4	0.0	-46.2	-46.1	0.0	-43.1	5.5	-9.3	5.1	-9.1	0.5	170	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.9	17.6	28.1	17.3	25.9	17.0	19.7	26.0	18.7	24.9	17.8	23.8	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	22.3	16.6	11.9	21.2	15.4	11.0	20.1	56.9	25.7	53.5	24.6	50.6	23.9	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.1	4.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-24.9	-18.8	-9.9	-1.9	4.3	15.6	17.9	13.8	6.6	-2.8	-17.7	-20.8		
	DBStd	16.19	10.61	8.81	8.21	5.64	5.10	5.30	3.77	3.74	4.38	5.86	9.83	10.34		
	HDD10.0	5384	1081	806	616	358	192	17	1	11	122	397	831	954		
	HDD18.3	7909	1339	1039	874	608	437	111	55	146	353	654	1081	1213		
	CDD10.0	590	0	0	0	0	14	184	245	128	19	1	0	0		
	CDD18.3	74	0	0	0	0	1	28	41	4	0	0	0	0		
	CDH23.3	1041	0	0	0	0	20	429	529	59	2	0	0	0		
	CDH26.7	304	0	0	0	0	5	139	158	2	0	0	0	0		
(14) Wind	WSAvg	1.8	1.5	1.7	2.0	2.3	2.4	1.9	1.8	1.6	1.7	1.9	1.7	1.6		
(15) Precipitation	PrecAvg	737	47	37	43	47	58	69	71	74	69	86	74	61		
	PrecMax	909	88	70	89	70	110	148	121	119	112	119	119	98		
	PrecMin	471	2	19	9	11	26	25	15	21	15	20	43	32		
	PrecStd	108	19	13	18	15	21	22	27	26	22	21	20	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.4	0.2	6.9	13.4	26.3	32.3	31.9	26.7	23.2	11.1	0.7	-0.7		
		MCWB	-2.7	-0.5	1.8	5.7	14.6	17.9	19.7	18.2	15.6	7.6	0.0	-1.4		
	2%	DB	-5.6	-1.9	4.1	10.3	19.7	30.4	30.0	25.1	19.0	7.6	-0.2	-2.5		
		MCWB	-6.3	-2.8	0.5	4.2	12.0	17.0	18.7	17.3	13.8	6.1	-0.8	-3.1		
	5%	DB	-7.3	-5.1	2.1	7.8	15.9	28.5	28.5	23.3	15.7	6.0	-1.7	-4.4		
		MCWB	-7.9	-5.8	-0.3	3.2	9.1	16.6	17.8	16.7	12.0	4.9	-2.4	-4.8		
	10%	DB	-10.1	-7.9	0.5	5.6	12.4	26.1	26.6	21.2	13.4	4.1	-3.0	-7.0		
		MCWB	-10.6	-8.6	-1.0	2.0	7.5	16.4	17.5	15.5	10.5	2.8	-3.7	-7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-0.3	2.2	6.5	15.1	19.8	22.0	20.3	16.0	8.3	0.2	-1.2		
		MCDB	-2.4	0.3	4.4	11.0	24.1	26.9	29.6	25.1	21.4	11.1	0.6	-0.5		
	2%	WB	-6.1	-2.8	1.0	4.8	12.8	18.7	20.5	18.4	14.2	6.4	-0.6	-3.1		
		MCDB	-5.5	-1.9	3.2	9.2	18.6	26.9	27.2	22.7	18.1	7.3	-0.2	-2.7		
	5%	WB	-8.0	-5.9	0.1	3.6	10.3	17.9	19.4	17.2	12.4	4.9	-2.2	-4.8		
		MCDB	-7.3	-5.1	1.6	7.2	14.4	25.5	25.5	21.7	15.5	6.0	-1.7	-4.4		
	10%	WB	-10.7	-8.5	-1.1	2.5	8.2	16.9	18.5	16.1	10.8	3.1	-3.7	-7.4		
		MCDB	-10.2	-7.9	0.7	5.3	12.0	23.9	24.3	20.5	13.1	3.9	-3.1	-7.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.6	11.8	10.0	9.3	12.5	11.4	10.5	7.7	5.8	9.3	8.8		
		MCDBR	7.8	9.1	11.1	13.4	15.6	17.8	15.7	14.5	11.8	7.0	7.1	7.4		
		MCWBR	7.5	8.8	7.9	8.4	8.0	6.8	6.3	7.5	6.7	5.6	6.9	7.4		
		MCDBR	7.7	9.4	9.1	11.2	13.7	14.0	12.8	12.4	10.8	7.2	7.3	7.3		
	5% WB	MCWBR	7.4	9.1	7.0	7.5	7.9	6.2	6.2	7.0	6.3	5.9	7.2	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.247	0.291	0.334	0.356	0.371	0.390	0.444	0.409	0.350	0.312	0.240	0.229		
	taud		1.919	2.006	1.987	2.048	2.267	2.362	2.182	2.260	2.401	2.255	1.969	1.876		
	Ebn at Noon		366	623	738	815	842	833	766	761	748	615	372	191		
	Edh at Noon		39	76	114	132	116	109	127	108	77	60	35	21		
(44) All-Sky Solar Radiation	RadAvg		0.28	1.14	2.64	4.35	5.06	5.59	4.95	3.42	2.04	0.83	0.35	0.13		
	RadStd		0.03	0.09	0.19	0.49	0.60	0.45	0.54	0.42	0.28	0.12	0.07	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page